T527

Multimedia Codec Benchmark

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History

Date	Author	Version	Description
2024.08.30	XAA0060	v1.1	add video/image decoder chapter
2024.09.10	AWA0508	v1.2	add encoder chapter
2024.10.09	XAA0060	v1.3	add decoder maximum bit rate
2024.10.09	AWA1438	v1.3	add decoder file format

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1. 前言

1.1. 适用范围

适用产品：T527、T527V、T527S

1.2. 性能测试场景说明

➢ 本文档所描述的 codec 的性能规格数据，是采用我司自研的测试程序（解码：android 平台的 Videos app，编码：EncoderTest）进行测试得出，作为性能基准供参考，而 codec 的性能受系统负载、物料差异、测试片源特性等因素的影响，不同的使用场景 codec 的性能可能会不一致，应以实测为准。

2. Video

2.1. Video Decoder

H.265	
Video Codec	H.265/HEVC Main/Main10 profile@Level5.1
Supported Image Size	32x32pixels to 8192x4352pixels
Maximum Frame Rate	4096x2304@60fps@10bit
Maximum Bit Rate	100Mbps
File Format	MP4, MOV, TS, MKV
Remark	Support Tile & Slice

VP9	
Video Codec	VP9 Profile0/Profile2
Supported Image Size	8x8pixels to 8192x8192pixels
Maximum Frame Rate	4096x2304@60fps@10bit
Maximum Bit Rate	3840x2160@100Mbps
	4096x2304@80Mbps
File Format	WEBM,MKV

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VP9	
Remark	

H.264	
Video Codec	H.264/AVC BP/MP/HP@ level 5.1
Supported Image Size	64x64pixels to 4096x2304pixels
Maximum Frame Rate	4096x2304@30fps@8bit
Maximum Bit Rate	3840x2160@100Mbps
	4096x2304@60Mbps
File Format	MP4, MOV, TS, FLV, AVI, MKV, 3GP
Remark	Support Field Coding Not support MBAFF Not support High 10 Profile

MPEG-1	
Video Codec	MPEG-1
Supported Image Size	16x16pixels to 1920x1088pixels
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	100Mbps
File Format	MP4,MOV,MKV,MPG,VOB,DAT
Remark	

MPEG-2	
Video Codec	MPEG-2 SP@ML,MP@HL
Supported Image Size	16x16pixels to 3840x2160pixels
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	100Mbps
File Format	TS,AVI, MP4,MOV,MPG, VOB,DAT
Remark	Support Field Coding

MPEG-4	
Video Codec	MPEG-4 SP@L0-3,ASP@L0-5，
Supported Image Size	16x16pixels to 3840x2160pixels

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MPEG-4	
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	100Mbps
File Format	MP4, MOV, 3GP,AVI, MKV
Remark	support GMC，support short header

H.263	
Video Codec	H.263 BP
Supported Image Size	16x16pixels to 1920x1088pixels
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	100Mbps
File Format	MP4,MOV,3GP,ASF
Remark	

VP8	
Video Codec	VP8
Supported Image Size	64x64pixels to 4096x2304pixels
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	40Mbps
File Format	WEBM,MKV
Remark	

AVS/AVS+	
Video Codec	AVS/AVS+ JiZhun
Supported Image Size	64x32pixels to 1920x1088pixels
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	30Mbps
File Format	TS
Remark	Support Field Coding

RMVB	
Video Codec	RMVB 8/9
Supported Image Size	64x64pixels to 1920x1080pixels

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RMVB	
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	10Mbps
File Format	RMVB
Remark	1.获得授权后提供支持补丁，可找 FAE 提供申请授权的方式

VC-1	
Video Codec	VC-1 SP@ML,MP@HL,AP@L0-4
Supported Image Size	128x96pixels to 2048x2048pixels
Maximum Frame Rate	1920x1080p@60fps
Maximum Bit Rate	100Mbps
File Format	MKV,TS,AVI,ASF,WMV
Remark	1.Support Field Coding 2.获得授权后提供支持补丁，可找 FAE 提供申请授权的方式

2.2. Video Encoder

H.264 encoder	
Video Codec	H.264/AVC BP/MP/HP@ level 5.1
Supported Image Size	144x48 pixels to 4096x4096 pixels
Maximum Frame Rate	3840x2160@25fps@8bit
Maximum Bit Rate	60Mbps
File Format	mp4, ts
Remark	Not support field encoding

3. Image

3.1. Image Decoder

JPEG	
Video Codec	JPEG
Supported Image Size	1x1pixels to 16384x16384pixels

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JPEG	
Maximum Data Rate	180M pixels/Second 1920x1080@60fps (YUV420)
Container	JPG, JPEG
Remark	Not support Non-interleaved Scan

3.2. Image Encoder

JPEG encoder	
Video Codec	JPEG
Supported Image Size	16x16 pixels to 8192x8192 pixels
Maximum Frame Rate	1920x1080@60fps
Maximum Bit Rate	60 Mbps
File Format	JFIF（JPEG File Interchange Format）
Remark	Not support Non-interleaved Scan

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